

Assignment05

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2026-05-10

Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic_Sampling_Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic_Sampling_Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data",
  ↪ "Aquatic_Sampling_Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))
```

Class code

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
```

```

        med_sal = median(salinity_ppt, na.rm = TRUE),
        med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↳ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality

```

```

left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪   = TRUE),
  site_full = fct_rev(site_full))

```

```

# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
  color = "red",
  size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
  y = "Median salinity (ppt)",
  title = "Site salinity") +
# different theme
theme_bw()

```

```

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# base layer
copepods <- ggplot(data = copepods,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
    fun = "median",
    size = 4,
    color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
    y = "log + 1 transformed average abundance/L",
    title = "Copepods") +
  # different theme
  theme_bw()

```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: different sites
- y-axis: log + 1 transformed average ostracod abundance per liter
- geometry to represent average abundance/L: `geom_jitter()`, circle points
- geometry to represent medians: `stat_summary()`, circle points with a distinct color

b. Wrangle your data

```
# create new object from cleaned aquatic invertebrate data
ostracods_df <- aq_ins_clean |>
  # keep only ostracod rows
  filter(taxon == "ostracod") |>
  # log + 1 transform average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows
ostracods_df |>
  slice_sample(n = 10)
```

A tibble: 10 x 24

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2023-03-03_NDC	2023-03-03 00:00:00	NDC	ostr~	5.73	7.45	0.688	5.64
2	2023-09-12_NVBR	2023-09-12 00:00:00	NVBR	ostr~	0	9.14	NA	20.3
3	2023-06-06_NPB1	2023-06-06 00:00:00	NPB1	ostr~	0.533	7.48	NA	2.38
4	2021-04-15_NPB	2021-04-15 00:00:00	NPB	ostr~	185.	8.09	3.57	8.24
5	2023-11-11_NEC	2023-11-11 00:00:00	NEC	ostr~	0	NA	38.6	0.25
6	2022-02-25_NEC	2022-02-25 00:00:00	NEC	ostr~	0	8.75	44.1	9.87
7	2023-02-21_NEC	2023-02-21 00:00:00	NEC	ostr~	0	8.56	NA	8.8
8	2020-02-21_NVBR	2020-02-21 00:00:00	NVBR	ostr~	4.8	7.34	4.1	6.1
9	2022-01-24_NEC	2022-01-24 00:00:00	NEC	ostr~	0.267	NA	NA	NA
10	2022-11-19_NPB1	2022-11-19 00:00:00	NPB1	ostr~	2	7.26	2.47	3.96

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```
# base layer
ostracods <- ggplot(data = ostracods_df,
  # x = site, y = log abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: individual survey points
```

```

geom_jitter(height = 0,
            width = 0.1,
            shape = 21,
            size = 2) +
# second layer: median per site, distinct in size and color
stat_summary(fun = "median",
            geom = "point",
            color = "deeppink",
            size = 5) +
# wrap long site labels for readability
scale_x_discrete(labels = label_wrap(15)) +
# readable axis labels and taxon-specific title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +
# non-default theme
theme_bw()

# display object
ostracods

```

